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Worlds Beyond Earth: Earth's Oceans

The Water Worlds Within Our World

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Introduction

Look at the Earth from any distance and the first fact announces itself: this is the water planet. More than seven-tenths of the surface lies under a connected body of salt water that holds ninety-seven percent of all the water in the world, a layer that averages nearly four kilometers deep and reaches almost eleven at its deepest soundings, and the whole assembly, the ocean basins, the currents, the chemistry, and the life within, is not decoration but machinery: the ocean stores and moves the sun's heat around the globe, breathes in partnership with the sky above it, feeds the clouds and the rains, swallows and gives back the carbon, and carries an interior world of ecosystems whose catalog is still unfinished. This book is about that machinery, told from the water's own point of view: how the ocean came to be, how it is built and moves, how it shapes the climate and the planet's habitability, and what the centuries of exploration have learned and have left to learn.

The story has a surprising opening: the ocean is not as old as the planet, but nearly. The Earth formed four and a half billion years ago from the swirl of dust and rock around the young sun, hot enough at first that any water on the surface would have boiled away, yet by some hundred million years later the rocks preserve the record of water standing and working at the surface, which means the ocean arrived early and has been on stage almost the whole time. A capsule on the isotope detective work belongs here, for the water's origin question is settled by chemistry's own fingerprints. The hydrogen of the sea carries a signature, the balance of the heavy and light isotopes, and the matching vaults read differently: the comets measured so far carry too much of the heavy hydrogen, the ocean's recipe not matching the icy messengers' own; the carbonaceous meteorites, the outer asteroid belt's own remnants, match the sea's signature closely, the rocky carriers being the better fit; and the mantle's own water, the mineral inclusions' sampled cargo, matches too, the planet's own vault reading the same, the capsule's point being the method's own elegance: the sea's origin is a forensic case, the water's own atoms testifying, the walker holding the isotope argument as the book's first detective story, the science's own favorite kind.

Where the water came from, whether it was baked out of the planet's own interior, delivered by icy bodies from beyond the frost line, or both, is one of the book's early questions, and the answer, like most of the ocean's answers, is more interesting than the simple version taught for decades.

The book's plan follows the water. The opening chapters set the stage: the ocean's origin, its inventory and structure, the shape of the basins that hold it, and the water's fundamental chemistry. The middle chapters are the machinery: the circulation that moves the currents, the waves and tides, the ocean's partnership with the atmosphere, its role in the climate machine, and the teeming life it carries from the sunlit surface to the alien hadal trenches. The later chapters are the human story: the centuries of sounding and diving and mapping, the modern satellite and robotic era, the open mysteries that remain, and the future of an ocean now being measured as it warms, rises, and acidifies under the weight of the civilization that studies it.

A promise on method before the voyage begins. The ocean is the best-measured large object on the planet and also, in its deep parts, one of the least explored; the difference between those two facts is exactly the difference between the surface and the deep, and this book keeps the distinction honest throughout, marking what is mapped, what is modeled, and what is still darkness on the chart. Whoever finishes this book will know the ocean as a system, its history, its engines, and its unsolved problems, and will know which is which, which is the only way to know the sea.

Chapter 1: Understanding Earth's Oceans

The ocean's story begins with the object itself, and this chapter supplies the working picture: what the ocean is, how it is counted, and how its parts are named.

A capsule on the ocean's own naming history belongs here, for the conventions carry the human story. The Atlantic's name descends from the Greek titan of the myth, the sea beyond the pillars of the known world; the Pacific's name is Magellan's own irony, the calm he met after the strait's terror earning the largest ocean its gentle name; the Indian's name is the trade route's own label; and the Southern's formal recognition, the hydrographers' 2000 decision, is the modern era's own honesty, the naming of the connector as a named body, the capsule's point being the walker's own map's age: the ocean's names are a museum of the sailing centuries, the geography's human layer, the book's standing reminder that the sea's map is a document as much as a nature.

The global ocean is one connected body of water; the names on the map, the Atlantic, the Pacific, the Indian, the

Arctic, and the Southern, are conventions of geography rather than boundaries of nature, water flowing freely between all of them around the storm-wracked south and through the deep passages of the great basins. The sea's own vertical statistics get a capsule here, for the deep water is the ocean's silent majority. The average depth's own census: the shallow shelf seas, the drowned edges, are a few percent of the ocean's area, the sunlit zone's own thin skin; the continental slopes and the abyssal plains, the four-to-six-kilometer country, hold the sea's own bulk, the abyssal plains alone covering more of the Earth than any other single environment, the planet's dominant landscape being an underwater flat no human has seen; and the trenches, the hadal realm, are a fraction of one percent of the floor, the pressure frontier's own rare real estate, the capsule's meaning: the ocean's true shape: the sea is a deep place, its average resident being a cold dark water over a sediment plain, the familiar beach being the exception's own edge, the audience holding the vertical census as the book's own standing perspective.

The Pacific alone, the largest by far, covers more area than all the continents combined and holds more than half the ocean's volume; the Arctic, the smallest and shallowest, sits over a drowned continent's fragments; and the Southern Ocean, the ring around Antarctica whose formal recognition came only in this century's geography, is the connector, the place where the water of every basin is stirred together, the ocean's true mixing bowl. A capsule on the comparison that calibrates the scale belongs here, for the ocean's size is best felt against the land. The ocean's own area is the planet's 71 percent, the land's 29 being the exception's own islands-and-continents arrangement; the sea's own average depth, the 3.7 kilometers, set against the land's average elevation of less than one, means the water's world is deeper than the dry world is high, the terrestrial half of the planet being the shallow part; and the human scale's own smallness, the deepest dive's own record being a few hours at the trench's floor while the surface world's whole civilization runs on the skin, being the standing proportion, the capsule's meaning: the audience's own corrected intuition: the Earth is an ocean with some dry high ground, not a land with some seas, the base map's own inversion being the book's first and lasting adjustment, the capsule's standing scale.

The sizes get their one-line census: the total area, some 361 million square kilometers, the volume, some 1.3 billion cubic kilometers, and the average depth, some 3.7 kilometers, numbers whose consequence, that the ocean is three hundred times the mass of the atmosphere, will drive half the chapters of this book.

The pressure's arithmetic gets a capsule in the ladder's context, for depth's physical meaning is the ocean's own ruler. The pressure rises one atmosphere per ten meters, the water's own weight, and the ladder's floors run: the sunlit zone's twenty atmospheres at its base, the midnight's hundred-plus, the trench's floor above a thousand, the pressure at the Challenger Deep being the weight of a hundred elephants balanced on a thumbnail, the standard comparison the field's own illustration; and the pressure's biological consequences, the deep life's own chemistry, the membranes and the proteins tuned to the crush, being the adaptation's own subject, the capsule's point being the ladder's honesty: each floor is not only darker but heavier, the ocean's depth being a dimension as real as altitude, the sea's own vertical mountain in reverse, the walker descending with the numbers in hand.

The water's anatomy is standard: five zones by light and depth, and the walker should hold the ladder. The sunlit zone, the top two hundred meters, is where the light drives the photosynthesis and where nearly all the ocean's visible life crowds; the twilight, down to a kilometer, holds the fading light and the nightly migration of the lanterned; the midnight, the largest habitat on the planet by volume, is the cold permanent dark of the deep; the abyssal, the plains at four to six kilometers, is the sediment's slow country; and the hadal, the trenches beyond six kilometers, named for the underworld's landlord, is the pressure-crushed frontier where the water's weight is a thousand times the air's, the whole ladder being the book's vertical map, every later chapter's events located on it.

The water's composition gets its capsule early, for the ocean is not pure water but a solution. The salt, averaging about 3.5 percent by weight, is mostly the familiar table salt's two elements, sodium and chloride, with magnesium, sulfate, calcium, and potassium in a supporting cast, the whole recipe nearly uniform across the globe, a fact the ocean's own mixing guarantees; and the salt's persistence, the ions arriving from the rocks and the rivers over billions of years and leaving only slowly in the evaporite basins and the vents' chemistry, is the reason the sea is salty while the rain that feeds it is fresh, the cycle's asymmetry being the salt's own history. The chapter's closing picture is the one to keep: a planet whose water layer, thin as an apple's skin at planetary scale but vast at human scale, is one connected chemical machine with one set of engines, the machine the rest of the book disassembles piece by piece.